# POWER BI PROJECT:

## *ANALYZE TEACHING ASSISTANT HELP SESSIONS AT KELLY*

# Prepared By:

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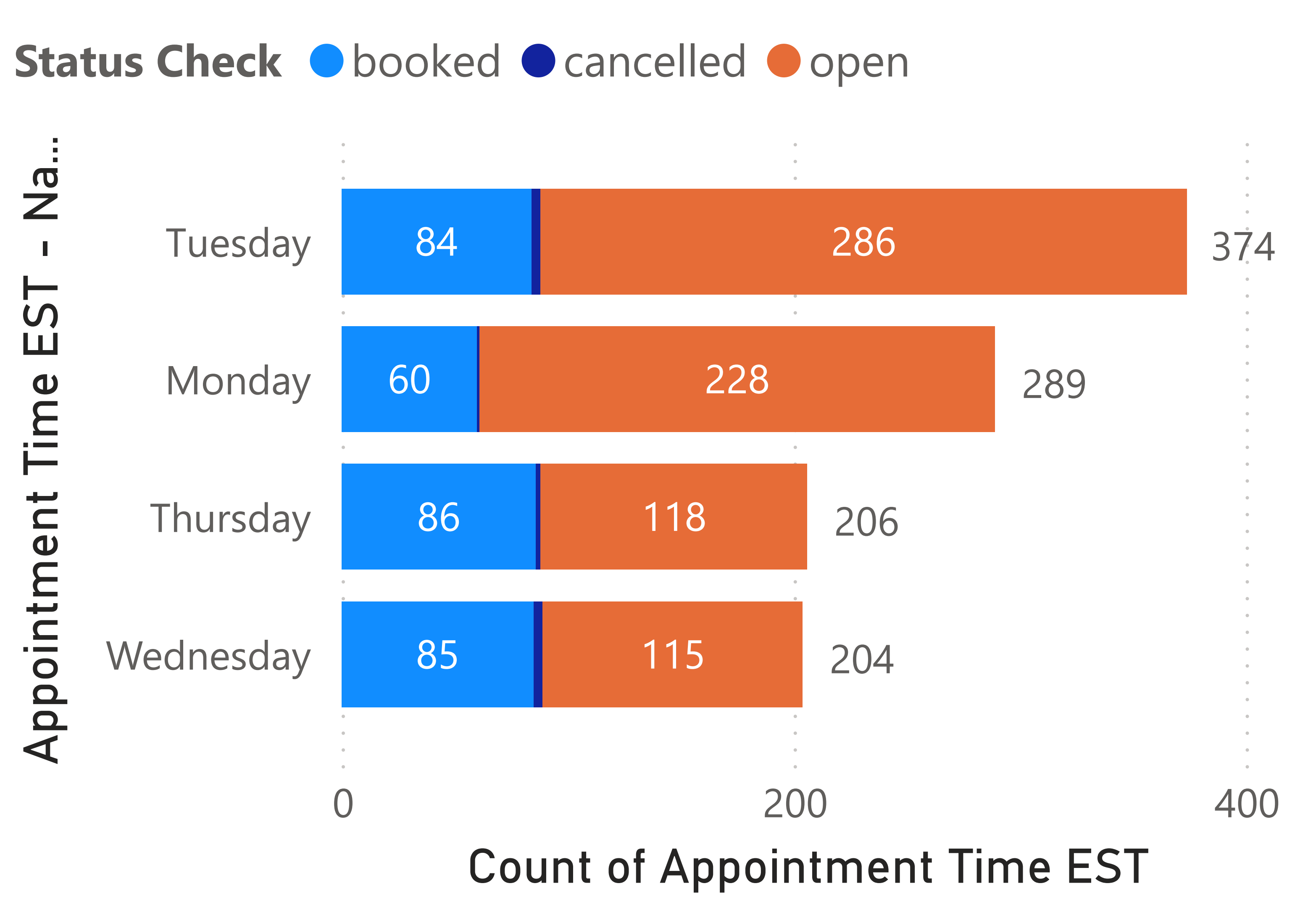
Final submission: https://app.powerbi.com/groups/2fc63a25-409e-49b3-9f52-409ec9a78d31/reports/6157337e-343e-4b71-be8c-f51e0829bac9/035713e748e40061c60c?experience=power-bi

Number of Bookings
326

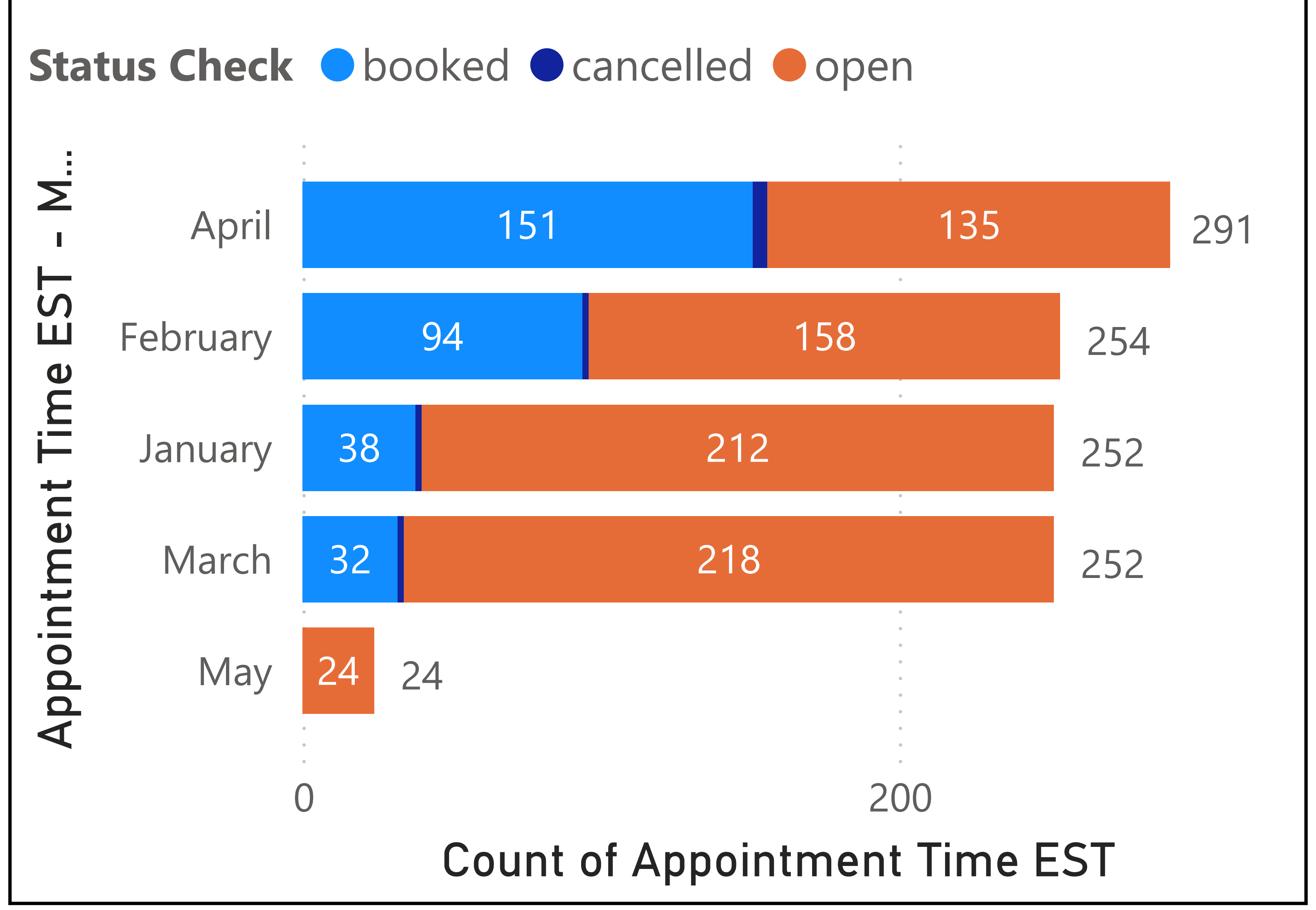
Number of Appointments
1.07K

Appointment Status
All

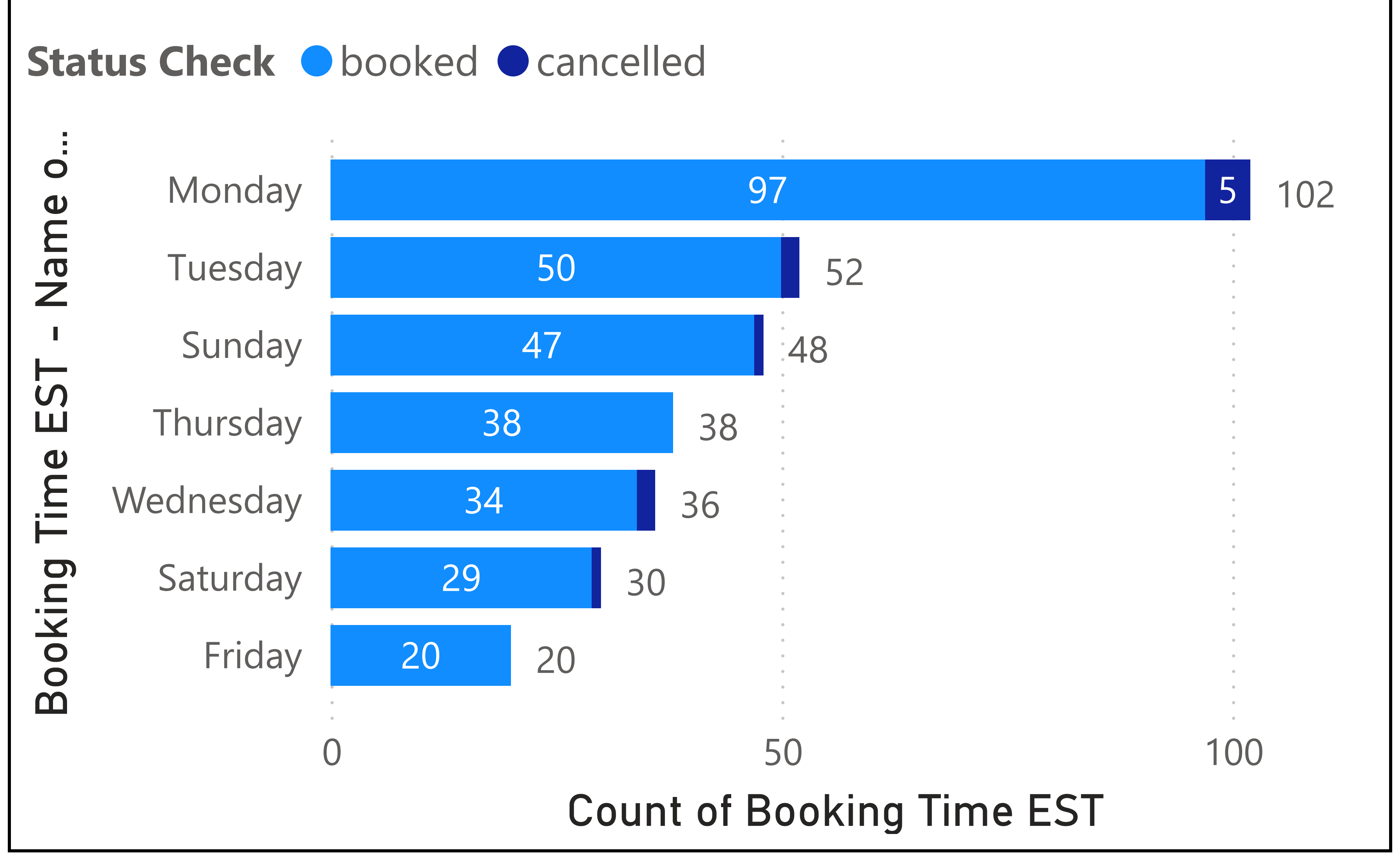
### Appointment Count By Day



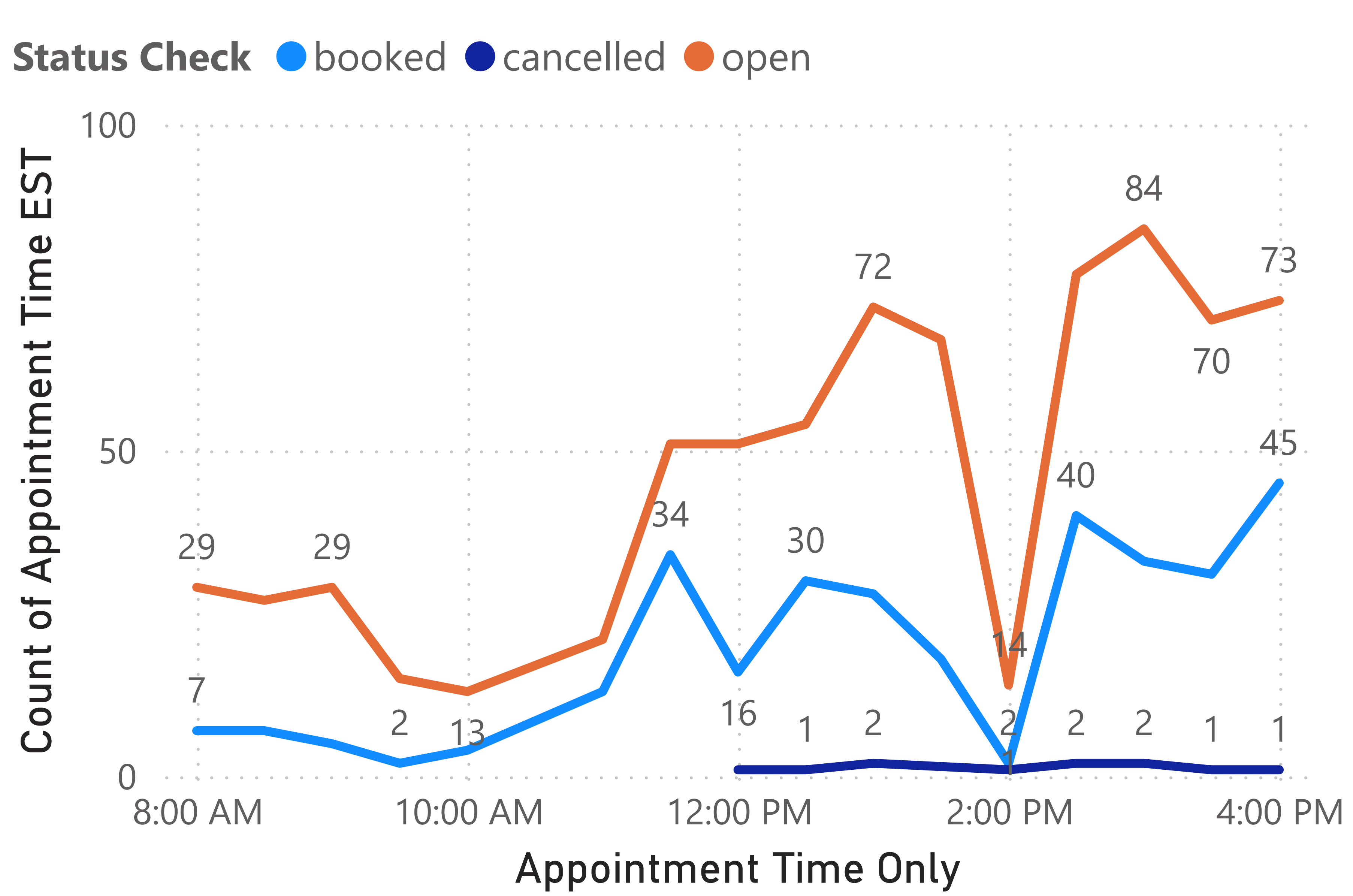

### Appointment Count By Month



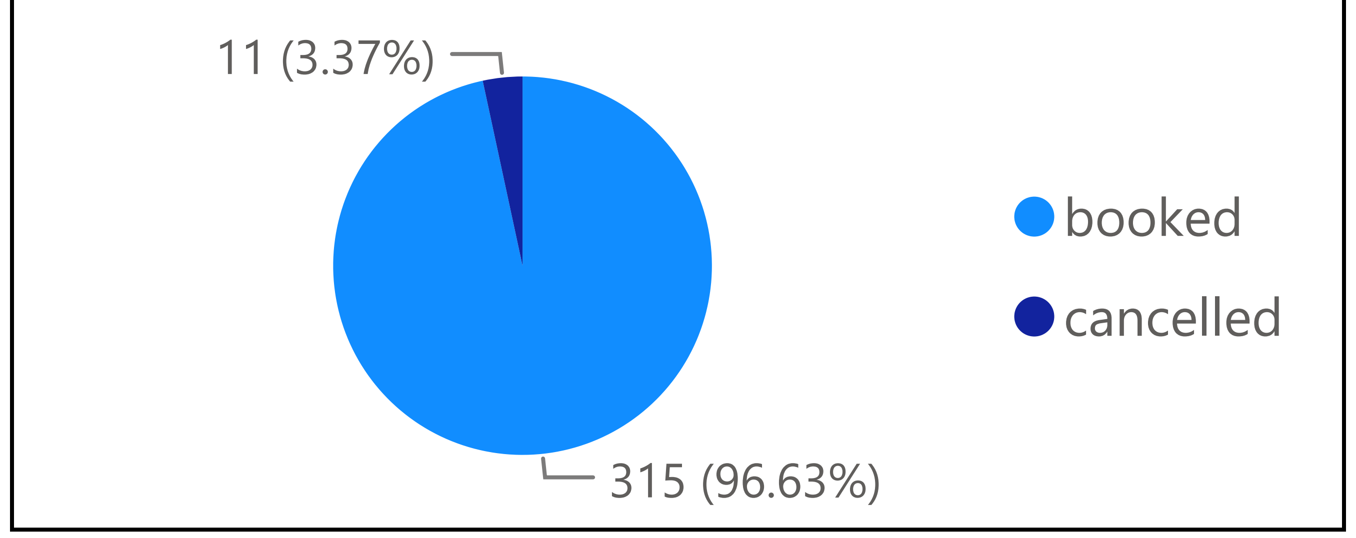

### Booking Count By Days



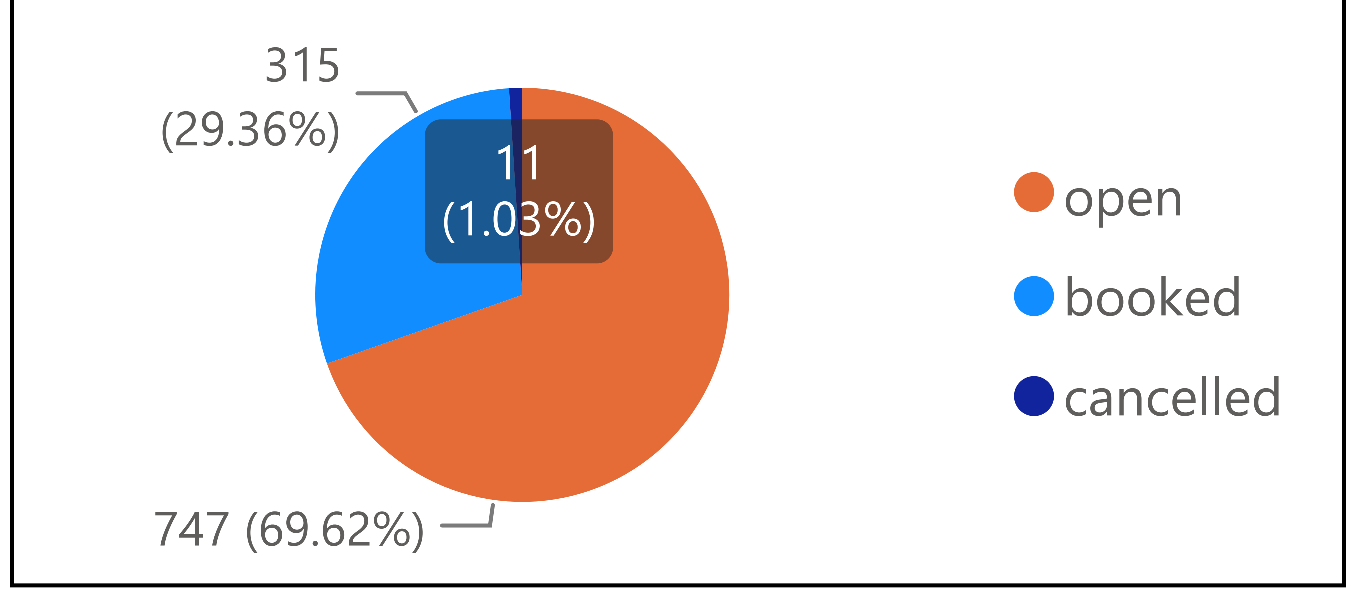

### Appointment Count By Hour



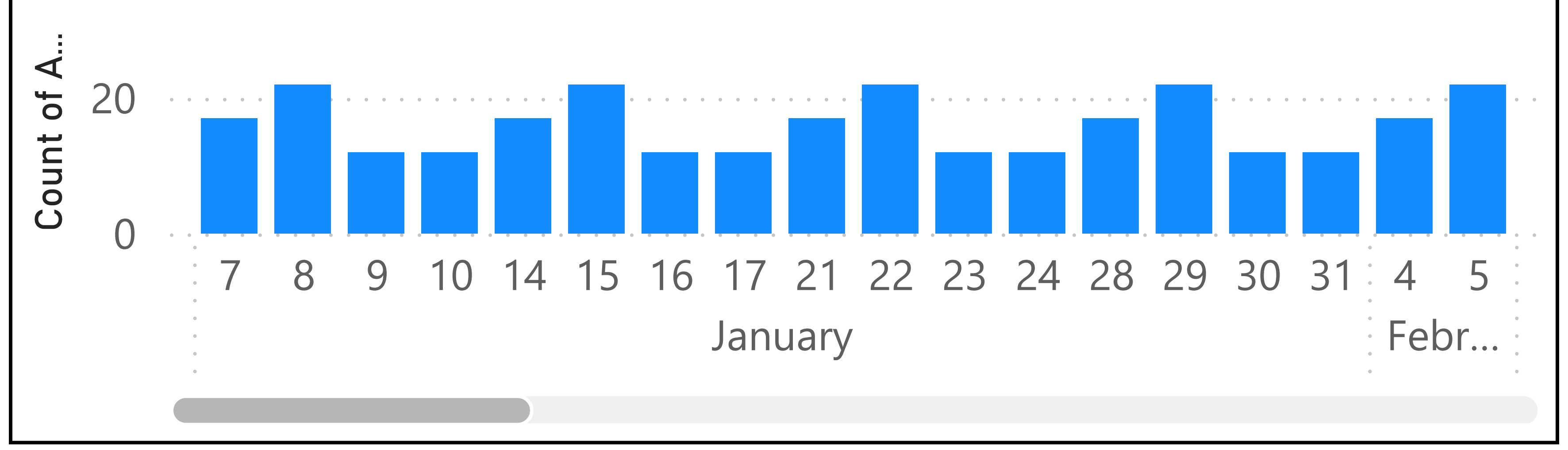

### % of Total Booking by Status



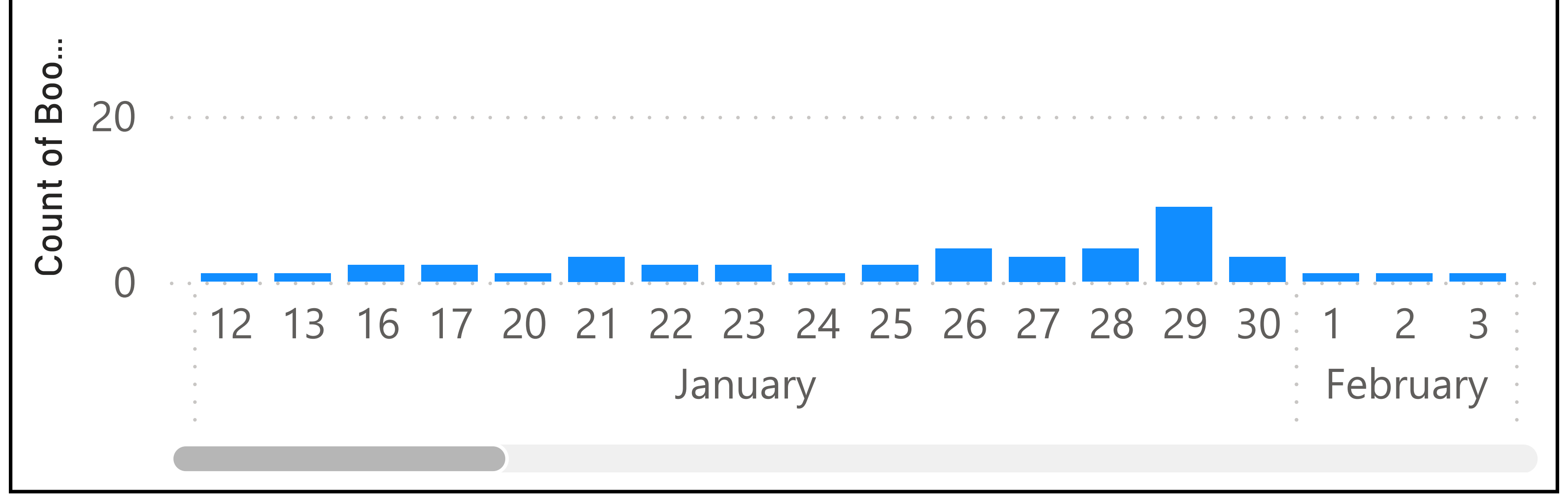

### % of Total Appointment By Status




### Count of Appointment ID by Month and Day




### Count of Booking Time EST by Month and Day

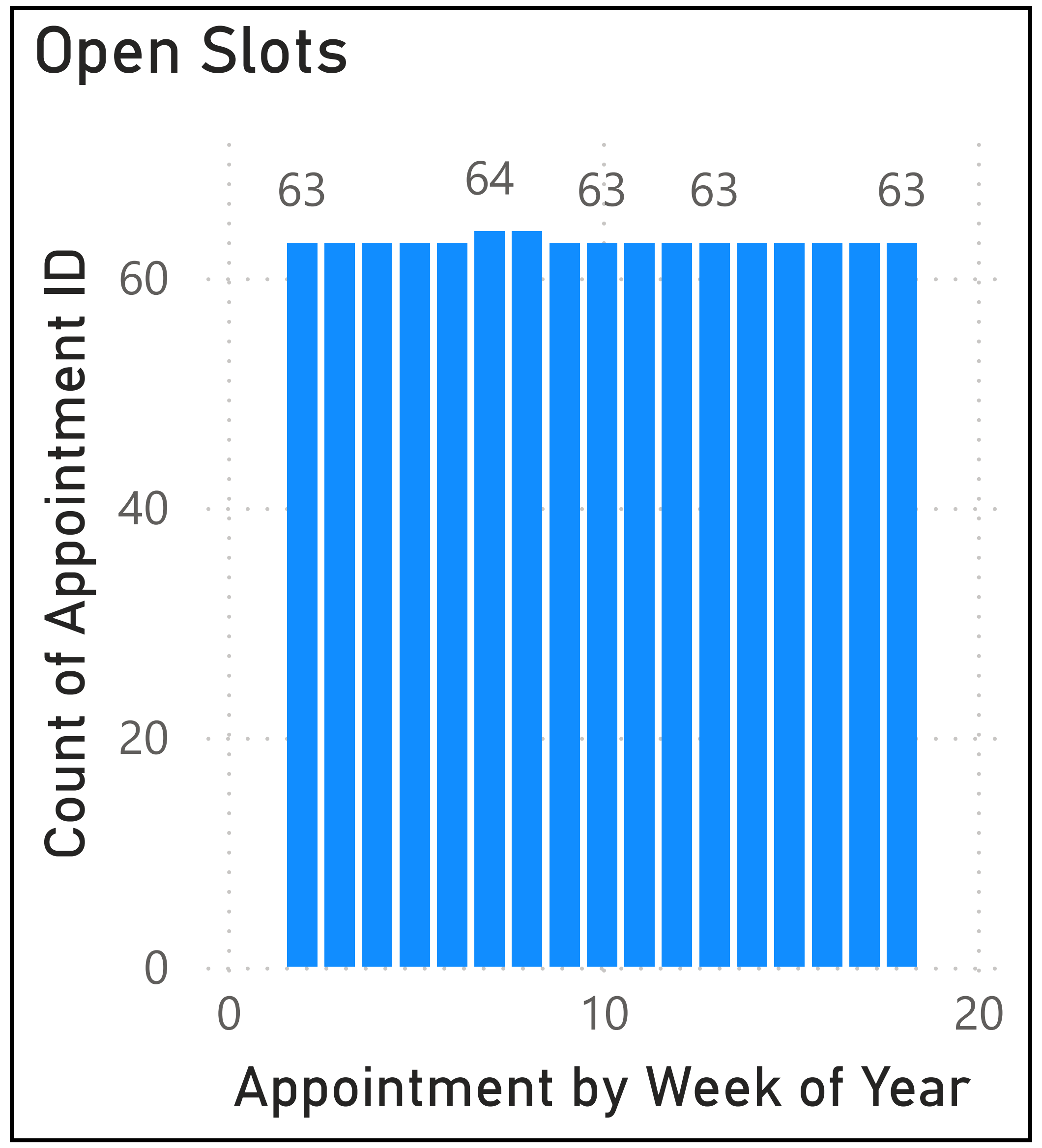


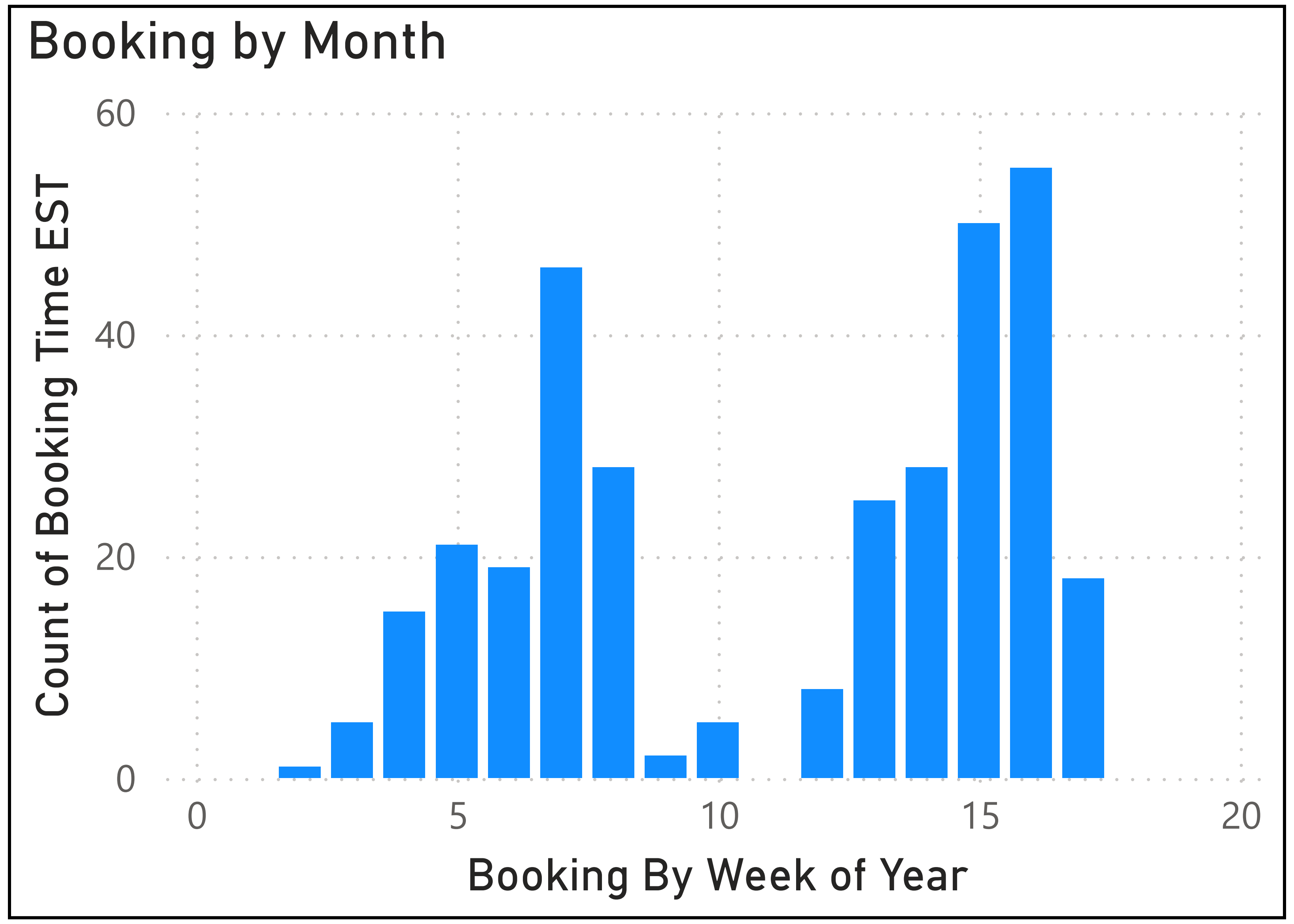
Status Check

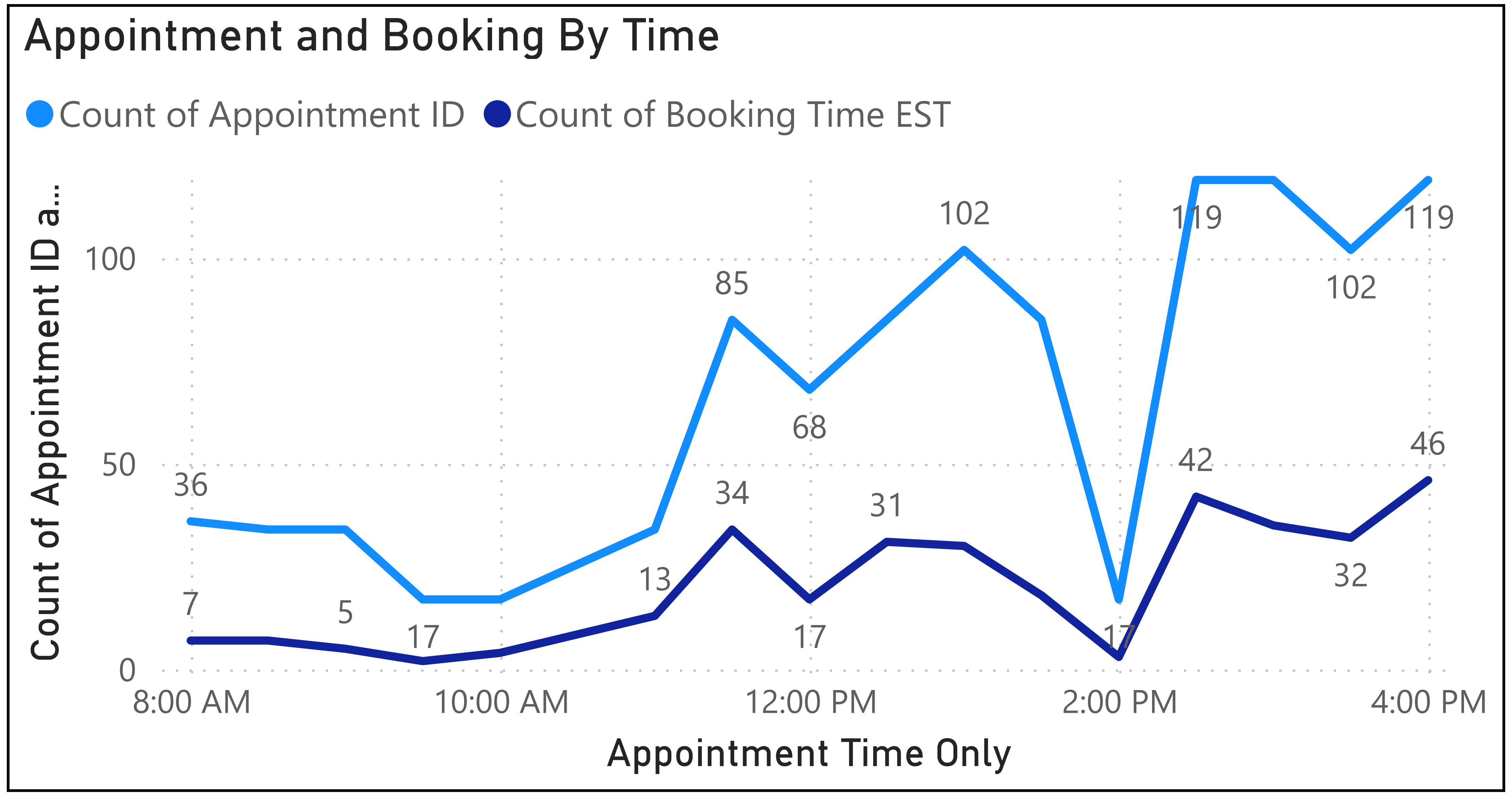
All

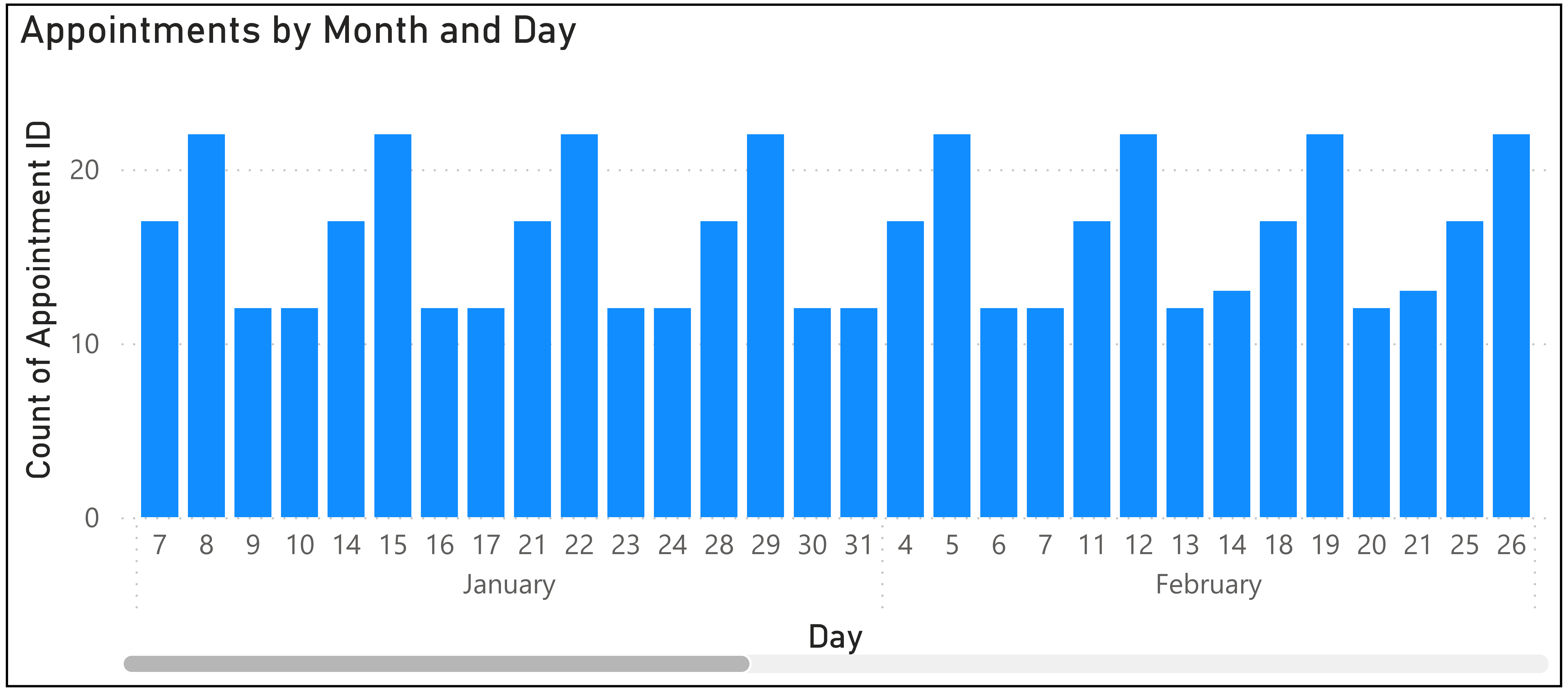
Lead Time (in Days)

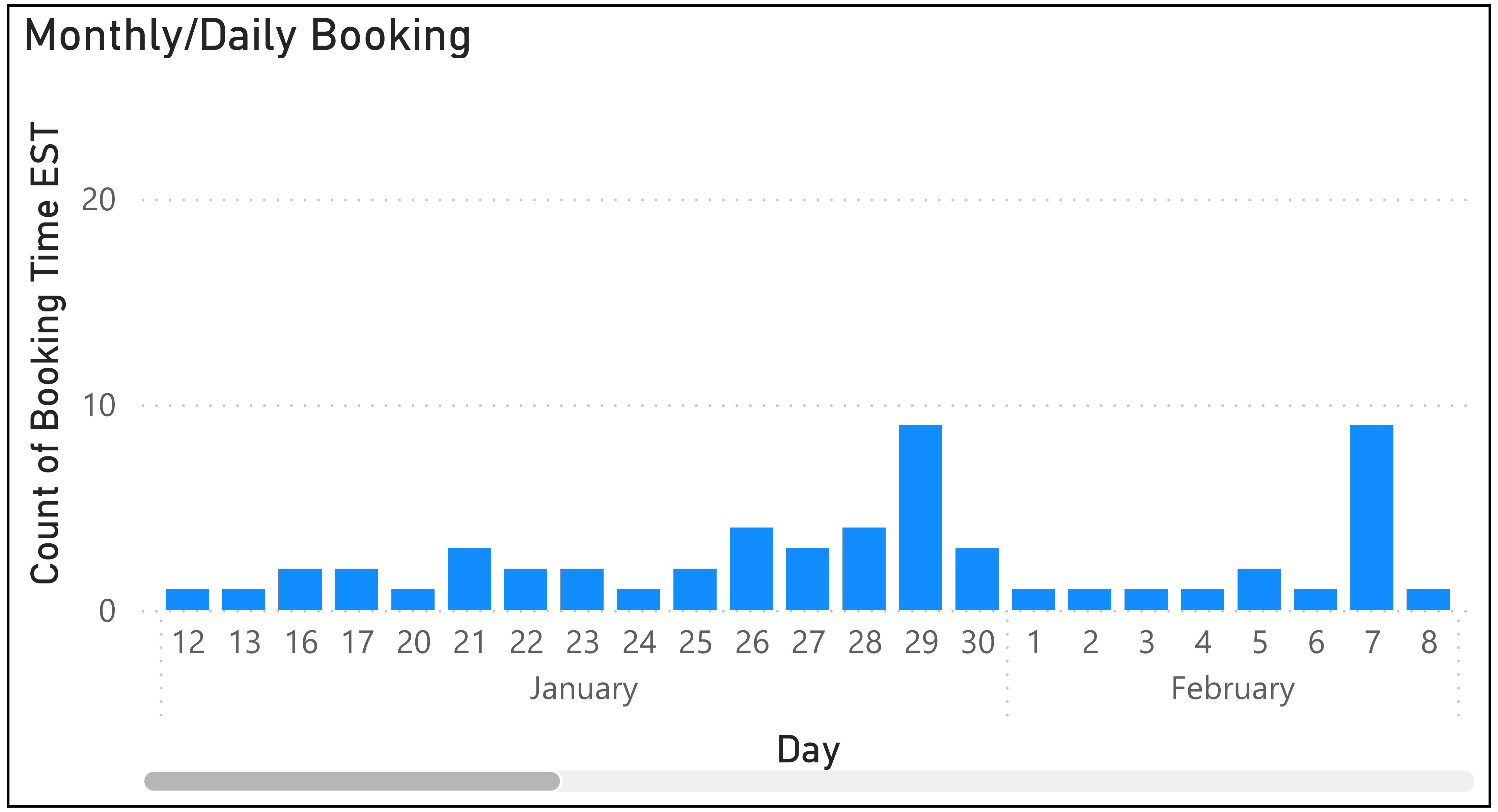
1.00 13.92



| | | | |
|---|---|---|---|
| **13.92** | **1.00** | **3.22** | **2.49** |
| Max lead time | Min Lead Time | Avg lead time | Median lead time |

Status Check

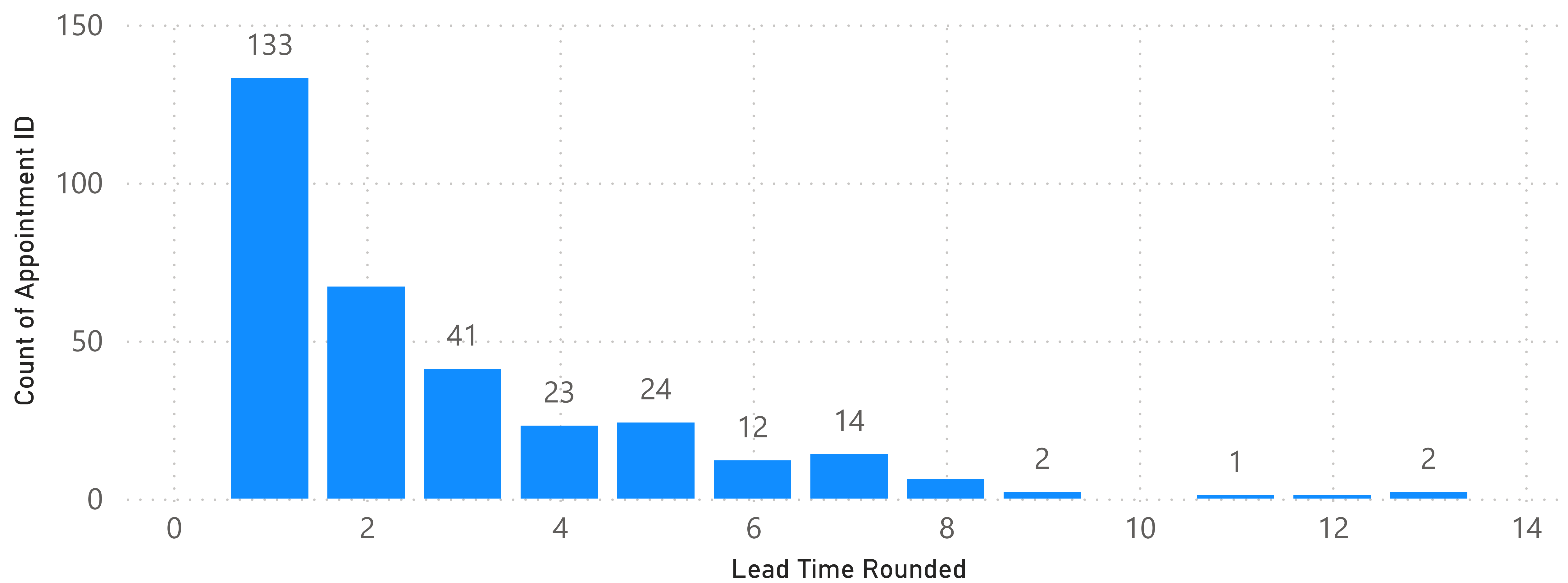
All

Quarter, Month, Day

All

How does lead time for appointment change overtime?


Count of Appointment By Lead Time Rounded

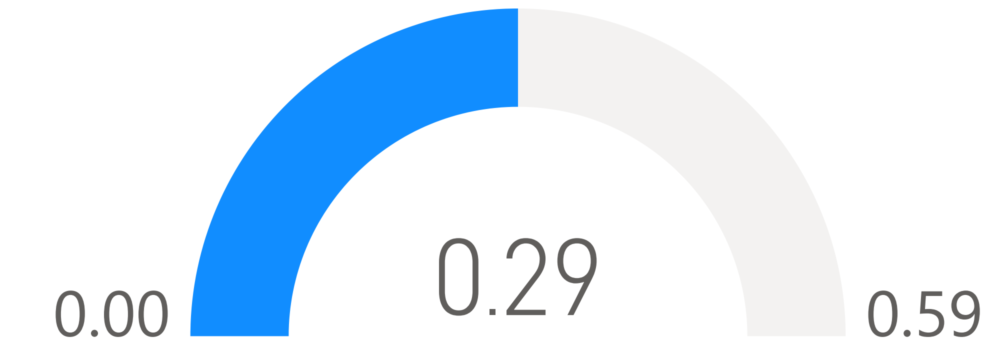


Lead Time (in Days)

1.00 | 13.92

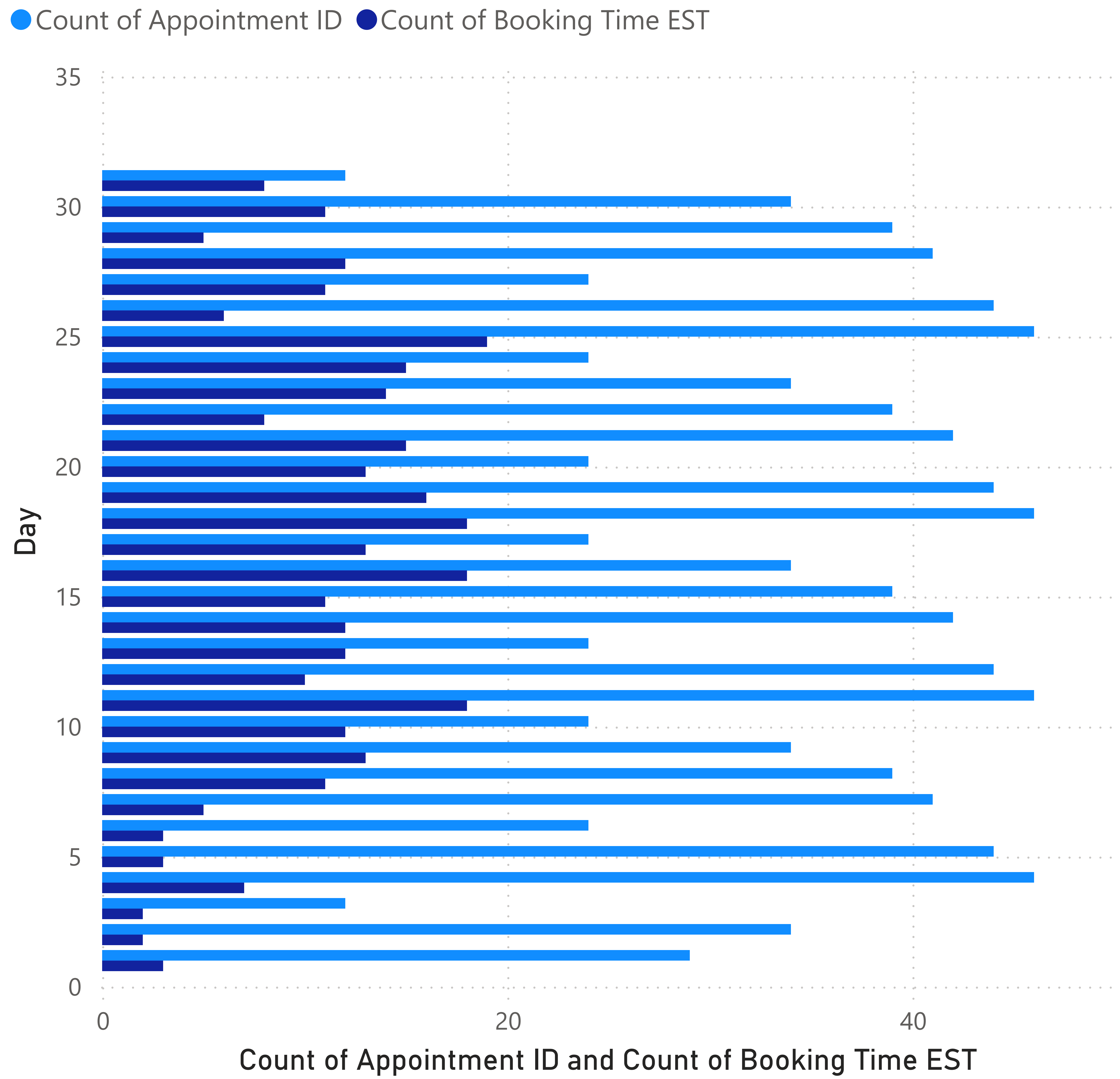
| | | | |
|---|---|---|---|
| **13.92** | **2.49** | **1.00** | **3.22** |
| Max lead time | Median lead time | Min Lead Time | Avg lead time |

Utilization




| TA | Utilization | Open Appointments | Booked Appointments | Cancelled Appointment |
|---|---|---|---|---|
| Angela Martin | 0.35 | 87 | 48 | |
| Anne Wentworth | 0.06 | 16 | 1 | |
| Bill Lumbergh | 0.18 | 27 | 6 | |
| Bob Porter | 0.21 | 26 | 7 | |
| Bob Slydell | 0.18 | 28 | 6 | |
| Brian Duffey | 0.41 | 38 | 28 | |
| Dom Portwood | 0.18 | 28 | 6 | |
| Drew Pitts | 0.18 | 28 | 6 | |
| Dwight Schrute | 0.50 | 16 | 17 | |
| Jim Halpert | 0.44 | 19 | 15 | |
| Joanna Aniston | 0.15 | 29 | 5 | |
| Kevin Malone | 0.14 | 116 | 20 | |
| Lawrence Bader | 0.21 | 27 | 7 | |
| Michael Bolton | 0.24 | 26 | 8 | |
| Michael Scott | 0.26 | 25 | 9 | |
| Milton Waddams | 0.26 | 25 | 9 | |
| Nina McInroe | 0.32 | 23 | 11 | |
| Pam Beesly | 0.35 | 21 | 12 | |
| Peter Gibbons | 0.32 | 23 | 11 | |
| Phyllis Lapin | 0.38 | 21 | 13 | |
| Ryan Howard | 0.41 | 19 | 14 | |
| Samir Nagheenanajar | 0.29 | 24 | 10 | |
| Stanley Hudson | 0.41 | 19 | 14 | |
| **Total** | **0.29** | **747** | **315** | **1** |

Appointment Date/Time By TA



Filter by month

Quarter, Month, Day

All

| Student ID | Count of Appointment ID | Booked Appointments | Cancelled Appointments | Open Appointments | Max lead time | Min Lead Time |
|---|---|---|---|---|---|---|
| | 747 | | | 747 | | |
| 100001 | 1 | 1 | | | 3.95 | 3.95 |
| 100002 | 10 | 8 | 2 | | 6.35 | 1.01 |
| 100003 | 26 | 21 | 5 | | 4.74 | 1.01 |
| 100004 | 4 | 4 | | | 6.51 | 1.08 |
| 100005 | 1 | 1 | | | 4.72 | 4.72 |
| 100006 | 5 | 5 | | | 6.09 | 1.51 |
| 100007 | 9 | 9 | | | 7.04 | 1.78 |
| 100008 | 4 | 4 | | | 1.64 | 1.07 |
| 100009 | 7 | 7 | | | 3.86 | 1.04 |
| 100010 | 2 | 2 | | | 7.65 | 5.66 |
| 100011 | 1 | 1 | | | 1.93 | 1.93 |
| 100012 | 3 | 3 | | | 4.96 | 1.48 |
| 100013 | 2 | 2 | | | 3.50 | 2.56 |
| 100014 | 3 | 3 | | | 2.51 | 1.84 |
| 100015 | 2 | 2 | | | 6.99 | 1.03 |
| 100016 | 6 | 6 | | | 7.11 | 1.67 |
| 100017 | 6 | 6 | | | 8.72 | 1.80 |
| 100018 | 10 | 6 | 4 | | 5.93 | 1.01 |
| 100019 | 9 | 9 | | | 13.77 | 1.00 |
| 100020 | 2 | 2 | | | 7.64 | 5.68 |
| 100021 | 4 | 4 | | | 7.97 | 1.06 |
| 100022 | 4 | 4 | | | 6.13 | 2.70 |
| 100023 | 2 | 2 | | | 5.76 | 1.56 |
| 100024 | 4 | 4 | | | 2.15 | 1.11 |
| **Total** | **1073** | **315** | **11** | **747** | **13.92** | **1.00** |

**2.97**
Avg Appointment Per Student

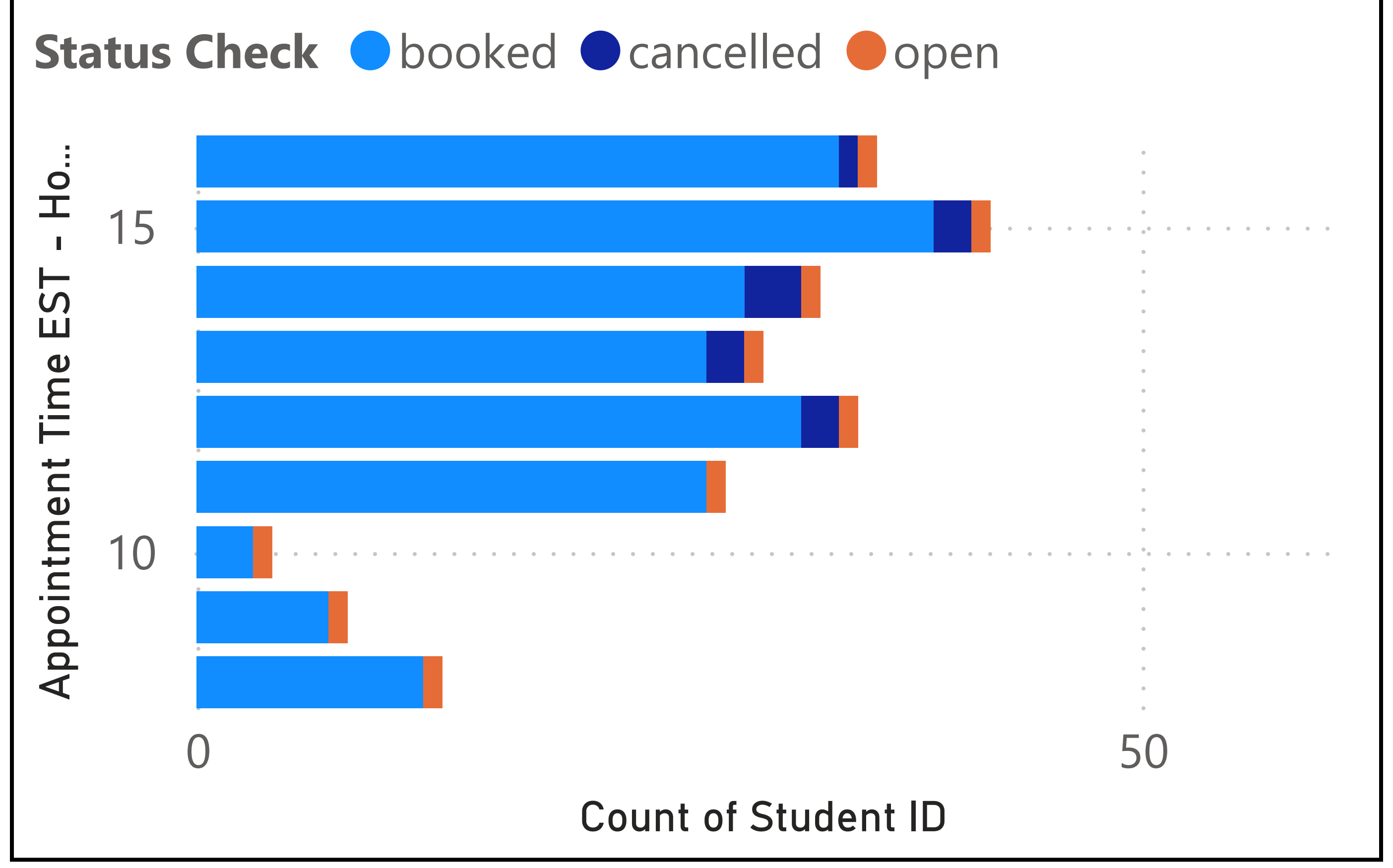
**106**
Total Student

**13.92**
Max lead time

**2.49**
Median lead time

**1.00**

Count of Student Appointment Day and Status Check



## INTRODUCTION

In this project you will assume the role of a teaching assistant coordinator with a large undergraduate course in Kelley. You will use Power BI Desktop to analyze a single semester of data from a web-based office hour system that teaching assistants have used in the past. You will connect to the data, transform it as you see fit, and visualize data to address concerns from the students and faculty as outlined below.

### OBJECTIVES

This case is designed to:

- Provide students an opportunity to utilize tools and concepts from the course in a case-based exercise.
- Facilitate the use of skills from the course to analyze data from a case.
- Assess your learning by answering questions related to the case utilizing Power BI Desktop reports.

This is an individually assigned project. You may not work with other students in the course (or prior students) on this project.

## CASE BACKGROUND

### OVERVIEW

You are working with a faculty member at Kelley. The issue at hand deals with a large course of undergraduate students (over 1,500 students per semester on average). This course offers regular help to all students in the form of teaching assistants (TA). Currently, each TA signs up to provide one-on-one office hours to assist students in the course. This system is maintained online, via an older web system that your faculty member did not design. They were provided only with a text output of the past semester of data (a spring semester). Your report does not need to consider any potential data from a fall semester. You have been briefed on the following:

- In theory the system sounds great. The idea of one-on-one assistance (tutoring) sounds good, but some logistical problems have arisen.
- TA's do not like system as they claim students either do not show or signup for time slots months in advance; effectively cutting off prime time slots from other students.
- Students do not like the system as they complain that TA's regularly are not at the scheduled times.
- Students also complain that the times available do not coincide with the due dates of projects in the course. Currently, major projects for the course are due on Sunday evenings.
- To complicate the situation more, space in the school is at a premium. Major renovations are about to take place and office space is already limited. The construction will further complicate this situation. Anything to free up space (i.e., be more efficient) would be welcomed.
- Budgets are also a concern. TA's in unoccupied office hours are wasting funds for the course that could be utilized elsewhere.

### ANALYSIS

You have been asked to analyze the available data from the scheduling system. Your faculty member would like to move to a "walk-in clinic" model of office hours. This approach is common in the medical community and the faculty member has been successful with such implementations prior to joining Kelley. The most significant obstacle is convincing colleagues that this method is effective. In their past experience, the faculty member has stated that many people are not only resistant to change in general but also distrust the walk-in clinic model.

Based on this, they would like your help developing a dashboard system to analyze the data. Any changes would need to be implemented in one or two semesters so there is plenty of time for data collection and analysis.

## GETTING STARTED

Due to technical and security issues, you cannot connect to the scheduling system directly. Instead, the system generates a TXT file periodically to a shared cloud storage folder. The file ***tblappointment.txt*** is stored in Canvas where you can download and connect to it using Power BI Desktop. Additionally, the **DataDictionary.xlsx** file located in Canvas can be used as a data dictionary for the TXT file provided above. It also contains a list of Teachings Assistants for the semester in question. Download the zip archive of files, unzip the folder before connecting to any data with Power BI Desktop. Inside you will find two files:

1. Tblappointment.txt
2. DataDictionary.xlsx

Before beginning your analysis with Power BI Desktop, build a data model that will connect all aspects of your data. Next, using the questions below construct Power BI dashboards to visualize the data and trends. The tblappointment.txt file contains 1,073 rows of data spanning the spring semester of 2019.

The data will require some transformations/cleansing. This can take place either in Power Query or in Model view. Of particular note, any date or times should be displayed in Eastern (Indiana) time. The native times from the scheduling system are in Unix time. See the DataDictionary.xlsx file in Canvas for specifics. Also, Googling “Unix Time to Excel” will provide you with some guidance about converting Unix time to Excel/Windows compatible times. Be careful to check your work on these steps. Your accuracy here is key to your success on this project (i.e. if your conversions are incorrect, all of your data insights will be incorrect as well).

To check your conversions consider the following examples:

***APPOINTMENT ID 5811 IS SCHEDULED FOR MONDAY, MARCH 11, 2019 AT 11:30 AM.***

***APPOINTMENT ID 5813 IS SCHEDULED FOR MONDAY, MARCH 25, 2019 AT 11:30 AM AND WAS BOOKED ON MARCH 23, 2019 AT 10:01:04 PM.***

For these time conversions keep in mind that you will need to account for time zone changes *and* daylight savings time. These conversions must be applied to both the appointment times and the booking times.

Feel free to add any additional columns you need for your analysis here. While you will be able to do this later in the project, Power Query is often an easy and convenient method of adding columns early in the project. Below you will perform significant analysis of time. Having fields for days of the week, week number, and hour of the day will be helpful in future analysis.